

NutriAI: Adaptive AI-Powered Nutritionist

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Abstract—The growing prevalence of obesity and other metabolic conditions necessitates the need for personalized and sustainable approaches to nutrition. Conventional food tracking apps are largely dependent on manual entries and calorie-based models that do not foster sustained adherence or personalization. NutriAI is a proposed adaptive system powered by artificial intelligence that incorporates behavioral and metabolic modeling, along with real-time feedback from the user. NutriAI uses data on physiological and behavioral signals, such as hunger, cravings, energy, and stress, to provide context-aware recommendations. NutriAI leverages a modern, full-stack architectural framework based on React, Next.js, and MongoDB technologies to minimize friction and enhance user experience.

Keywords—Machine Learning, Next.js, MongoDB, TfidfVectorizer, Nutrition Tracking, Behavior Analytics, Cosine Similarity, Natural Language Processing (NLP), Linear Regression

I. INTRODUCTION

The increasing frequency of lifestyle-related health conditions, including obesity and metabolic disorders, has highlighted the need for more effective and accessible nutritional guidance. Traditional approaches to diet tracking and weight management often rely on manual logging and generalized recommendations, which can be difficult to maintain and may not align with individual needs. As a result, many individuals struggle to stay consistent, achieve sustainable weight loss, and develop long-term healthy habits.

This paper introduces NutriAI: Adaptive Nutrition Tracking Application, a system designed to simplify diet monitoring and support healthier lifestyle choices. The application allows users to track their daily food intake in an organized and efficient manner while providing structured feedback based on their goals and habits. By focusing on ease of use and practical functionality, NutriAI aims to reduce the challenges commonly associated with maintaining a balanced diet.

Unlike conventional diet-tracking methods, NutriAI emphasizes adaptability and user engagement. The system is designed to help users

better understand their eating patterns and make informed decisions without adding unnecessary complexity. By streamlining the tracking process and presenting relevant insights, the application supports users in building consistent routines and improving overall health.

The objective of this paper is to present the design, features, and potential benefits of NutriAI, as well as to evaluate its effectiveness as a tool for diet management and weight loss. Through a practical and user-focused approach, NutriAI aims to contribute to more sustainable and accessible nutrition tracking solutions.

II. CURRENT SOLUTIONS

Several existing applications address diet tracking and weight management, each using a different approach but sharing similar limitations. MyFitnessPal is one of the most widely used platforms, allowing users to log meals, track calories, and monitor exercise through a large food database; however, it relies heavily on manual input, which can be time-consuming and may reduce long-term consistency [1]. Lose It! also focuses on calorie tracking and personalized weight-loss goals, offering features such as barcode scanning and structured targets, but it still depends on users regularly entering their data [2]. In contrast, Noom takes a different approach by incorporating psychology-based lessons to help users build healthier habits over time, although it often requires a greater time commitment and paid subscription [3]. Overall, while these applications provide useful tools for managing diet and weight, they highlight common challenges such as the effort required for tracking and maintaining long-term engagement.

III. PRODUCT REQUIREMENTS

The product requirements for NutriAI: Adaptive Nutrition Tracking Application explain the main features and system qualities needed, supporting effective diet tracking, measuring food intake and promoting healthier lifestyles.

Consumers using the App can create and manage a personal profile, using personal information like sex, age, height, weight, and fitness goals to monitor diet as part of a personal health management program. The application allows users to log their daily food intake using a searchable food database. It will then calculate and display caloric intake, along with basic nutritional information. In addition, the system will track user progress, including changes in caloric intake and weight, plus graphically show the user's progress toward their goals. The app will also generate simple summaries that help users understand their eating habits, allowing them to manage food and hunger better.

The non-functional requirements explain how the system will perform. The app will be easy to use, with a simple and clear interface that makes daily tracking straightforward. It will respond to user inputs quickly, ensuring a smooth experience. All User Data will be stored securely using the latest best practices, protecting privacy. The system will be designed to be reliable, with minimal downtime, and should be scalable, easily handling an increasing number of users. It should also be compatible with mobile devices for maximum interoperability and accessibility.

Overall, these requirements are the clear foundation for designing and developing NutriAI as a practical tool for navigating dietary choices in the modern world and improving overall health.

IV. USER PROFILE AND PERSONALIZATION FLOW

Application users build and manage a detailed personal profile, including information like medical history, fitness level, and diet restrictions. This data is used to calculate key metrics, including Body Mass Index (BMI) and estimate activity energy levels in real time. This information helps personalize the user experience and provide a foundation for adaptive recommendations in the app. By capturing both lifestyle and physiological

inputs, the application is able to align its feedback with user needs better.

V. GOAL SETTING

Users are able to define their personal health and fitness goals, including target weight and overall wellness objectives. The system uses profile data, like BMI and current weight, to build goals and provide direction to the user for progress tracking. In this way, users can clearly understand their starting point and monitor their changes over time. The application encourages consistent behavior and promotes long-term habit formation by connecting user goals with their daily inputs.

VI. HUNGER AND CRAVING INTERACTION

The application also includes interactive features allowing users to report how they feel throughout the day. This "hunger check" evaluates user experiences using simple scoring logic to determine whether the user is physically hungry or craving food. Based on this, the system guides users toward the best choices, like healthy meal suggestions or alternative behaviors that manage cravings. We think this approach helps users become more aware of their eating habits, thus helping them make more informed decisions.

VII. FEATURE NAVIGATION FLOW

The application has a centralized Features Page, allowing users access to key tools like hunger or craving interactions, and of course, the tracked diet information and progress over time. Based on user behavior, the system routes users to different pages, providing a guided and responsive experience. This structured navigation guides decision-making and reduces the need for manual user input. By organizing features in a clear and accessible manner, the application improves the user experience and makes it more seamless.

VIII. METHODOLOGIES

NutriAI uses a client-server architecture designed to support real-time user interaction, health tracking, and AI powered recipe recommendations. The system is made up of four components: the frontend web application, a backend API layer, a database, and a custom made AI service.

A. Frontend

The frontend interface of NutriAI was built using the Shadcn component library and Tailwind CSS in order to create a maintainable component-based architecture. Unlike traditional component libraries that had strict styling constraints, Shadcn provides unstyled primitives that are directly integrated into the application. This allows them to be fully controlled at the application level rather than having them treated as an external dependency. This allows the system to maintain both a consistent design and gives the developer flexibility in customizing the components' structure and behavior to meet the application requirements.

Tailwind CSS is used to help style the Shadcn components. Tailwind CSS is a utility-first framework that allows for precise control over the layout, space, and typography of HTML elements. This eliminates the need to have separate CSS files and reduces the need for context switching during development [6].

The combination of Shadcn and Tailwind CSS allows for a modular user interface, where complex interfaces are made up of smaller, reusable blocks such as cards, buttons, forms, etc. For example, daily logs, dialogs, and week selectors are implemented as modular components that share consistent styling and allow for components to be reused wherever in the application.

This frontend styling and component architecture allows for scalability, efficiency, and a seamless user experience.

B. Backend API

The backend of NutriAI was designed using a RESTful API architecture in order to provide a consistent and scalable interface between the frontend application and the data layer. The API routes are structured around the main resources such as daily logs, meals, user data, etc. This allows the system to handle requests in a predictable and organized manner with each route responsible for handling a specific operation such as creating, retrieving, updating, or delete data.

In order to ensure consistency and reduce repetition across all endpoints, a reusable route handler pattern was implemented. This centralizes common functionality such as request parsing, input

validation, authentication, and error handling. Abstracting these functions, this minimizes code duplication and ensures that all API routes behave the same.

Input validation is handled using a schema validator, like Zod. This ensures that all incoming data meets the required format before any processing is done. This prevents any invalid data from being stored in the database. In addition, responses also follow a standardized format so that both successful and error responses follow a consistent structure, making it easier for the frontend to handle different scenarios.

C. AI API (FastAPI)

Considering the AI Recommendation Model is coded in Python, and the frontend uses Next.js and TypeScript for app development, an API was necessary to integrate the two coding languages. Python is a versatile scripting language that provides many libraries for model creation, and is the most commonly used programming language when working with AI or Machine Learning models because of this.

FastAPI was used to bridge this connection, serving as the link between the main application and the AI Recommendation logic for the “I’m Hungry” and “I’m Craving” features. The process begins when a user selects either “I’m Hungry” or “I’m Craving” on the frontend, triggering a request to the API with the user’s input. The recommendation logic in the API then processes the user input and searches the database for valid recipes to recommend to the user. After finding a valid recipe, the database returns the result to the AI Recommender to finalize its logic and return a response in JSON format. This response is then displayed on the frontend, in which the user can choose to either generate a new response or accept the recommendation and end the cycle.

D. AI Model

The AI model is used to interact with the applications “I’m Hungry” and “I’m Craving” features. It is meant to generate a recipe that fits the user’s calories and macronutrients for the day, generating a recipe that fits these constraints when they’re “Hungry”, and allowing the user to input text when they have a specific “Craving”.

The AI recommendation logic involves Natural Language Processing (NLP) techniques by incorporating text-based feature extraction using TF-IDF (Term Frequency - Inverse Document Frequency), implemented through Scikit-learn's `TfidfVectorizer`. TF-IDF vectorizes terms by calculating the Term's Frequency (TF), the Inverse Document Frequency (IDF), and multiplying the TF and IDF scores to result in a weight for each term. The formulas for Term Frequency (TF) and Inverse Document Frequency (IDF) can be seen below [11]:

$$TF = \frac{\# \text{ of time the term appears in the document}}{\text{total \# of terms in the document}}$$

$$IDF = \log\left(\frac{\# \text{ of documents in the corpus}}{\# \text{ of documents in the corpus contain the term}}\right)$$

TF-IDF vectorization assigns weights to words in a sentence, ultimately performing a feature extraction for important keywords. This sorts out filler words, and then a Cosine Similarity function is used to compare these vectorized terms with terms in the desired search field. Relationships are established between these terms, ultimately returning a similarity score to define which terms match the closest.

Additionally, the AI recommendation logic includes filters and recipe classification to improve the accuracy of the AI Recommender. This aids in distinguishing irrelevant results, as there are cases where the terms may score high in Cosine Similarity due to keyword overlap, but are not a correct recommendation for the context of the app.

A logistic regression model is also proposed as a future enhancement to learn from the user's accepted/rejected recipes and their preferred foods from their daily log. This will allow the system functionality to grow from recommending random recipes that fit the user's calories and macronutrients to recommending insightful recipes by learning patterns in the user's behavior.

E. Data Handling

NutriAI uses MongoDB as its primary database, with Mongoose used for schema modeling and data interaction. A document-based database was chosen in order to support flexible and dynamic data

structures, which are well-suited for storing user health data and daily activity logs [7].

The core data model is centered around a daily log, where each user has a single document tracking all of their health data for a given day. This data includes fields such as weight, sleep, stress, and energy, as well as arrays for meals, hunger events, and craving events. Grouping all the related data into a single document, it simplifies the system and makes data retrieval easier since it reduces the need for complex queries.

Mongoose is used to define schemas and enforce structure within the database. This ensures that all the data stored follows a consistent format while still allowing for flexibility in the way data is organized [8].

F. Deployment

NutriAI is deployed using Next.js on Vercel in order to take advantage of the tightly integrated, serverless environment [9]. Because Next.js is developed by Vercel, the framework is optimized for deployment on the platform, allowing both the frontend and backend to be managed by a single system. This reduces the complexity of configuring separate hosting solutions for different parts of the application.

A serverless deployment model was chosen in order to eliminate the need for managing dedicated servers and to allow the system to scale automatically based on demand [10]. Because NutriAI involves user interactions, like logging data and interacting with AI, traffic patterns can vary day to day. Serverless functions ensure that resources are allocated dynamically, allowing for better efficiency while also maintaining performance. This also reduces the need for manual configuration and ongoing system maintenance, which allows development to remain focused on the application's functionality rather than deployment complexities.

IX. RESULTS

The development and testing of NutriAI: Adaptive Nutrition Tracking Application showed that the system is effective in helping users consistently track their diet and better understand their eating habits. During testing, we were able to

log meals quickly using the searchable food database, which reduced the time required compared to traditional manual tracking methods. Most users were able to complete daily entries without difficulty, indicating that the interface is simple and easy to use.

The calorie and macronutrient calculations were accurate based on the input data, and we also saw that the displayed information helped us make more informed food choices throughout the day. The progress tracking feature also proved useful, as we could clearly see weight changes and how closely they were meeting their goals. This visibility encouraged more consistent engagement with the application.

The AI recipe recommender proved to be effective in recommending recipes based on a specified user input. Enhancements had to be made to improve the AI's contextual understanding of the project. For example, it would correctly identify items that had some correlation with the word "dinner", while also recommending items like "dinner rolls" which logically would not be considered a meal. After defining some constraints and rules for calories and meal types, the AI recommender's accuracy improved immensely, being able to recommend recipes for implicit data.

In addition, the system maintained stable performance during testing. The application responded quickly to inputs, and no major issues with crashes or data loss were observed. The system was also able to handle multiple users without noticeable slowdowns, showing that it can scale effectively for broader use.

X. CONCLUSIONS

This paper presented the design and implementation of NutriAI: Adaptive Nutrition Tracking Application, a system created to simplify diet tracking and support healthier lifestyle choices. The results demonstrate that the application successfully meets its main objectives by providing an easy way to log food, track calories, and monitor progress over time.

The system addresses common challenges found in existing diet tracking solutions, particularly the difficulty of maintaining consistency and understanding personal eating habits. By offering a

straightforward interface and clear feedback, NutriAI helps users stay engaged and make better decisions regarding their diet.

Overall, NutriAI is an effective and practical tool for diet management and weight loss. Future improvements could include expanding the food database, adding more detailed nutritional insights, and improving cross-platform functionality to further enhance user experience and accessibility.

REFERENCES

- [1] "A Guide to Food Tracking App," *Fitness Together*, 2024. <https://fitnesstogether.com/centralgeorgetown/blog/a-guide-to-food-tracking-app> (accessed Apr. 11, 2026).
- [2] "5 Apps That Can Help You Track Your Food and Exercise: Novell Primary Care and Weight Loss: Weight Loss Clinic," *Novellpcp.com*, 2024. <https://www.novellpcp.com/blog/5-apps-that-can-help-you-track-your-food-and-exercise>
- [3] S. McLean, "8 Best Calorie Counter Apps (2024) | Garage Gym Reviews," *Garage Gym Reviews*, Nov. 20, 2024. <https://www.garagegymreviews.com/best-calorie-counter-apps>
- [4] E. Landes, "The 12 Best Weight Loss Apps of 2023," *Healthline*, Jan. 23, 2023. <https://www.healthline.com/nutrition/best-weight-loss-apps>
- [5] C. Snyder, "The 7 Best Weight Loss Apps of 2026: Approved by Experts," *Fortune.com*, Mar. 13, 2026. <https://fortune.com/article/best-weight-loss-apps/> (accessed Apr. 11, 2026).
- [6] tailwindcss, "Tailwind CSS," *tailwindcss.com*, 2025. <https://tailwindcss.com/>
- [7] MongoDB, "What Is MongoDB?," *MongoDB*, 2024. <https://www.mongodb.com/company/what-is-mongodb>
- [8] M. D. Team, "Tutorial: Mongoose Get Started," *Mongodb.com*, 2025. <https://www.mongodb.com/docs/drivers/node/current/integrations/mongoose/mongoose-get-started/>
- [9] "Vercel Functions," *vercel.com*, Mar. 06, 2026. <https://vercel.com/docs/functions>
- [10] "Next.js on Vercel," *Vercel*. <https://vercel.com/frameworks/nextjs>
- [11] Karabiber, F. (n.d.). TF-IDF — term frequency-inverse document frequency—LearnDataSci. <https://www.learnatasci.com/glossary/tf-idf-term-frequency-inverse-document-frequency/>